

Week 5-6 Exercise Evidence

Name: Jake Van Slyke GitHub repository URL:

https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter

Part A - Starting validation

- Local pytest result: 15 passed, 1 deprecation warning, using Python 3.13.11 and pytest 8.4.1 in the saved screenshot.
- Local test command: `python -m pytest -v`.
- Initial CI workflow run URL:
https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/actions/runs/37495623958
- Initial CI commit: `ddfd6e30531e6638e2f2912def40f2a27ce5f521` on `main`.

```

SDI4213 Weeks5_6 GitHub Starter  main*  sdi4213-week56
> python -m pytest -v
===== test session starts =====
platform linux -- Python 3.13.11, pytest-8.4.1, pluggy-1.6.0 -- /home/jake/school/devops/weeks_5-6/SDI4213_Weeks5_6_GitHub_Starter/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/jake/school/devops/weeks_5-6/SDI4213_Weeks5_6_GitHub_Starter
configfile: pyproject.toml
testpaths: tests
plugins: anyio-4.15.1
collected 15 items

tests/test_api.py::test_root_route_returns_message PASSED [ 6%]
tests/test_api.py::test_health_check_returns_ok PASSED [ 13%]
tests/test_api.py::test_read_items_returns_count_and_items PASSED [ 20%]
tests/test_api.py::test_read_item_returns_one_item PASSED [ 26%]
tests/test_api.py::test_read_item_returns_404_for_missing_item PASSED [ 33%]
tests/test_api.py::test_add_item_creates_item PASSED [ 40%]
tests/test_api.py::test_add_item_rejects_duplicate_id PASSED [ 46%]
tests/test_services.py::test_get_all_items_returns_starter_items PASSED [ 53%]
tests/test_services.py::test_get_item_by_id_returns_matching_item PASSED [ 60%]
tests/test_services.py::test_get_item_by_id_returns_none_for_missing_item PASSED [ 66%]
tests/test_services.py::test_calculate_total_quantity PASSED [ 73%]
tests/test_services.py::test_is_low_stock_true_when_quantity_at_threshold PASSED [ 80%]
tests/test_services.py::test_is_low_stock_false_when_quantity_above_threshold PASSED [ 86%]
tests/test_services.py::test_create_item_adds_new_item PASSED [ 93%]
tests/test_services.py::test_create_item_rejects_duplicate_id PASSED [100%]

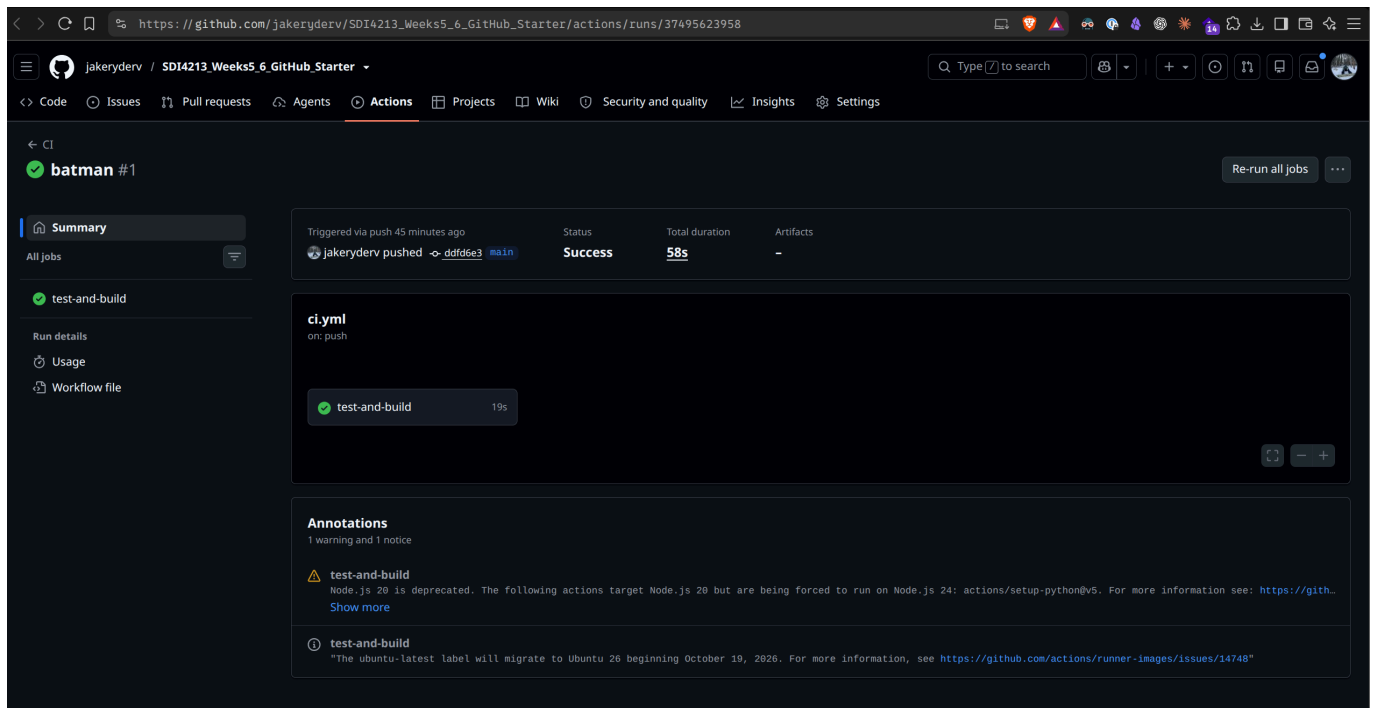
===== warnings summary =====
.venv/lib/python3.13/site-packages/starlette/testclient.py:51
/home/jake/school/devops/weeks_5-6/SDI4213_Weeks5_6_GitHub_Starter/.venv/lib/python3.13/site-packages/starlette/testclient.py:51: DeprecationWarning:
The anyio.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
  _PortalFactoryType = Callable[[], AbstractContextManager[anyio.BlockingPortal]]

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 15 passed, 1 warning in 0.15s =====

```

Milestone 1 - All 15 starter tests pass locally with the uv setup.

The local screenshot already uses `pyproject.toml`; it records local success after setup changes rather than an untouched-project baseline.



Milestone 2 - Baseline GitHub Actions CI workflow passes.

Workflow run:

https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/actions/runs/37495623958

Setup validation

This project uses uv to manage its Python environment and dependencies. The standard `pyproject.toml` file centralizes project metadata, dependencies, and pytest/Ruff configuration, while `uv.lock` records resolved dependency versions. Make commands provide consistent tooling commands locally and in CI. A generated `requirements.txt` preserves compatibility with the assignment's Dockerfile and ZIP package.

- Setup issue: https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/issues/1
- Requirements export: `make requirements` completed successfully on 2026-10-06.
- Local checks: `make check` completed successfully on 2026-10-06: Ruff lint passed, all 16 Python files were already formatted, and all 15 tests passed with 1 dependency deprecation warning.
- Warning: Starlette's test client references the deprecated `anyio.abc.BlockingPortal` alias. The warning did not fail the tests.

Part B - Week 5 build automation

- Build issue: https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/issues/3
- Pull request URL: https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/pull/4
- Successful workflow run URL:
https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/actions/runs/37510119164
- Build commit: `c416ea0636616ae3a54ff3b4819202450e241727` on `feature/build-artifact`.
- Artifact name: `sdi4213-app`.
- Packaged ZIP: `sdi4213-app.zip` inside the downloaded GitHub Actions artifact archive.
- Downloaded package contents: `app/`, `requirements.txt`, `README.md`, and `VERSION`. The application directory contains `__init__.py`, `main.py`, `models.py`, and `services.py`.

- Verification: GitHub Actions CI passed on 2026-10-06 and uploaded one artifact. The downloaded artifact was extracted locally, and its application ZIP contained all four required entries without Python cache files.

The screenshot shows the GitHub Actions interface for a workflow named 'Add automated build artifact #5'. The run was triggered via a pull request 20 minutes ago, has a status of 'Success', a total duration of 22s, and 1 artifact. The workflow file is 'ci.yml' and the job is 'test-and-build', which took 19s. The 'Annotations' section shows a warning about Node.js 20 being deprecated and a notice about the ubuntu-latest label migration. The 'Artifacts' section lists the uploaded artifact 'sdi4213-app' with a size of 10.3 KB and a SHA256 digest.

Name	Size	Digest
sdi4213-app	10.3 KB	sha256:216687e9a42dc7c8b53f8121f383753b3f7b368c6adc46...

Milestone 3 - The GitHub Actions run succeeds and lists the uploaded **sdi4213-app** artifact.

Downloaded artifact extraction

After downloading the artifact into `~/tmp/devops-weeks56`, extract the GitHub Actions archive, then the application ZIP it contains:

```
cd ~/tmp/devops-weeks56
unzip sdi4213-app.zip -d test-extract
unzip test-extract/sdi4213-app.zip -d test-extract/package
tree test-extract/package
```

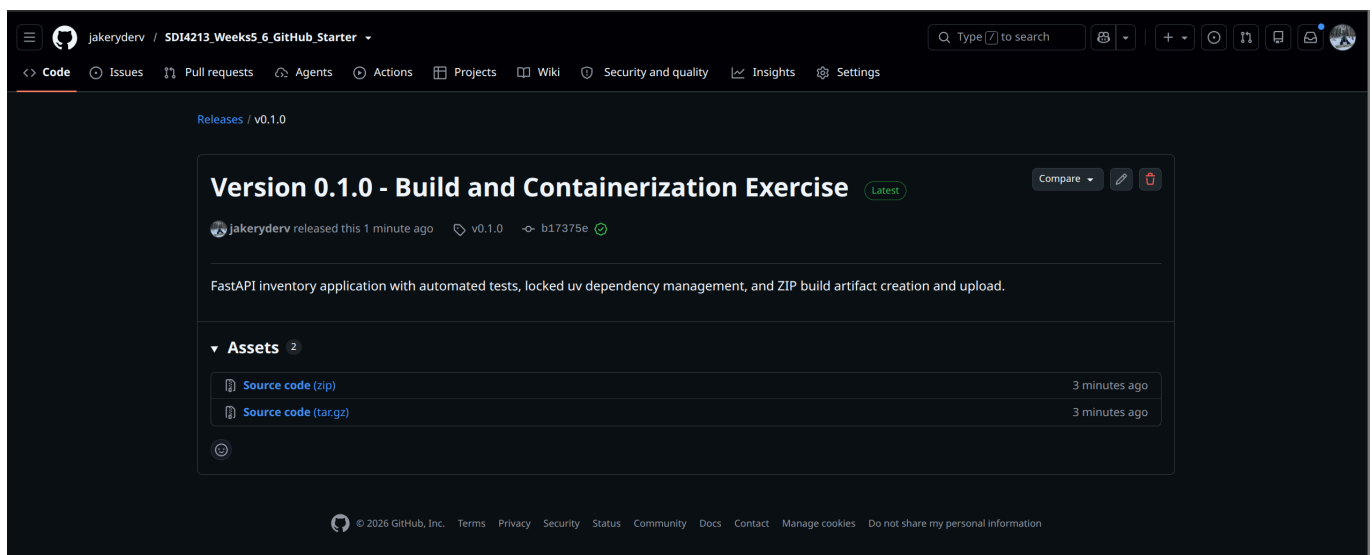
```
~/tmp/devops-weeks56
> tree test-extract/package
test-extract/package
├── app
│   ├── __init__.py
│   ├── main.py
│   ├── models.py
│   └── services.py
├── README.md
├── requirements.txt
└── VERSION

2 directories, 7 files
```

Milestone 4 - The terminal listing of the extracted package shows `app/`, `requirements.txt`, `README.md`, and `VERSION`, with 2 directories and 7 files.

Part C - Version and release

- Version: `0.1.0`.
- Git tag: `v0.1.0`.
- GitHub Release URL:
https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/releases/tag/v0.1.0
- Release title: `Version 0.1.0 - Build and Containerization Exercise`.
- Short release-note summary: FastAPI inventory application with automated tests, locked uv dependency management, and ZIP build artifact creation and upload.



Milestone 5 - Versioned GitHub Release v0.1.0 published.

Part D - Week 6 Docker

- Containerization issue: https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/issues/5
- Working branch: `feature/docker-container`.

Docker setup verification

- `docker --version`: Docker version 29.8.2, build `7fc2dff`.
- `docker compose version`: Docker Compose version `v5.6.0`.
- `docker run --rm hello-world`: completed successfully on 2026-10-06 with the `Hello from Docker!` message; the verification container was automatically removed.

Application container evidence

- Docker image name and tag: `sdi4213-week56:0.1.0`.
- Image ID: `b80580f7f030`.
- Container name: `sdi4213-week56-demo`.
- `docker images` evidence: Screenshot 6 shows the versioned image using `docker images | grep 'sdi4213-week56:0.1.0'`.
- `docker ps` evidence: Screenshot 7 shows container `d8b42395196c` running from `sdi4213-week56:0.1.0`, with host port 8000 mapped to container port 8000 for IPv4 and IPv6.
- `/health` response: `curl -fsS http://localhost:8000/health` returned `{"status":"ok"}` in Screenshot 8.
- `docker logs` evidence: Screenshot 9 shows application startup completing, Uvicorn listening on `0.0.0.0:8000`, and `GET /health HTTP/1.1` returning `200 OK`.

Build and run commands:

```
docker build -t sdi4213-week56:0.1.0 .
docker run -d -p 8000:8000 \
  --name sdi4213-week56-demo \
  sdi4213-week56:0.1.0
```

```
SDI4213_Weeks5_6_GitHub_Starter feature/docker-container*
> docker images | grep 'sdi4213-week56:0.1.0'
sdi4213-week56:0.1.0
```

```
b80580f7f030      300MB      73.6MB
```

Milestone 6 - Versioned Docker image built successfully.

Command: `docker images | grep 'sdi4213-week56:0.1.0'`.

```
SDI4213_Weeks5_6_GitHub_Starter feature/docker-container* zsh
> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
d8b42395196c   sdi4213-week56:0.1.0   "python -m uvicorn a..." 4 seconds ago  Up 4 seconds  0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp  sdi4213-week56-demo
```

Milestone 7 - Container is running and port 8000 is published.

Command: `docker ps`.

```
SDI4213_Weeks5_6_GitHub_Starter feature/docker-container*  
> curl -fsS http://localhost:8000/health  
{"status":"ok"}%
```

Milestone 8 - Containerized application health check succeeds.

Command: `curl -fsS http://localhost:8000/health`. Response: `{"status":"ok"}`.

```
SDI4213_Weeks5_6_GitHub_Starter feature/docker-container*  
> docker logs sdi4213-week56-demo  
INFO:      Started server process [1]  
INFO:      Waiting for application startup.  
INFO:      Application startup complete.  
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)  
INFO:      172.17.0.1:38796 - "GET /health HTTP/1.1" 200 OK
```

Milestone 9 - Container logs confirm the application started and handled requests.

Command: `docker logs sdi4213-week56-demo`.

Container cleanup

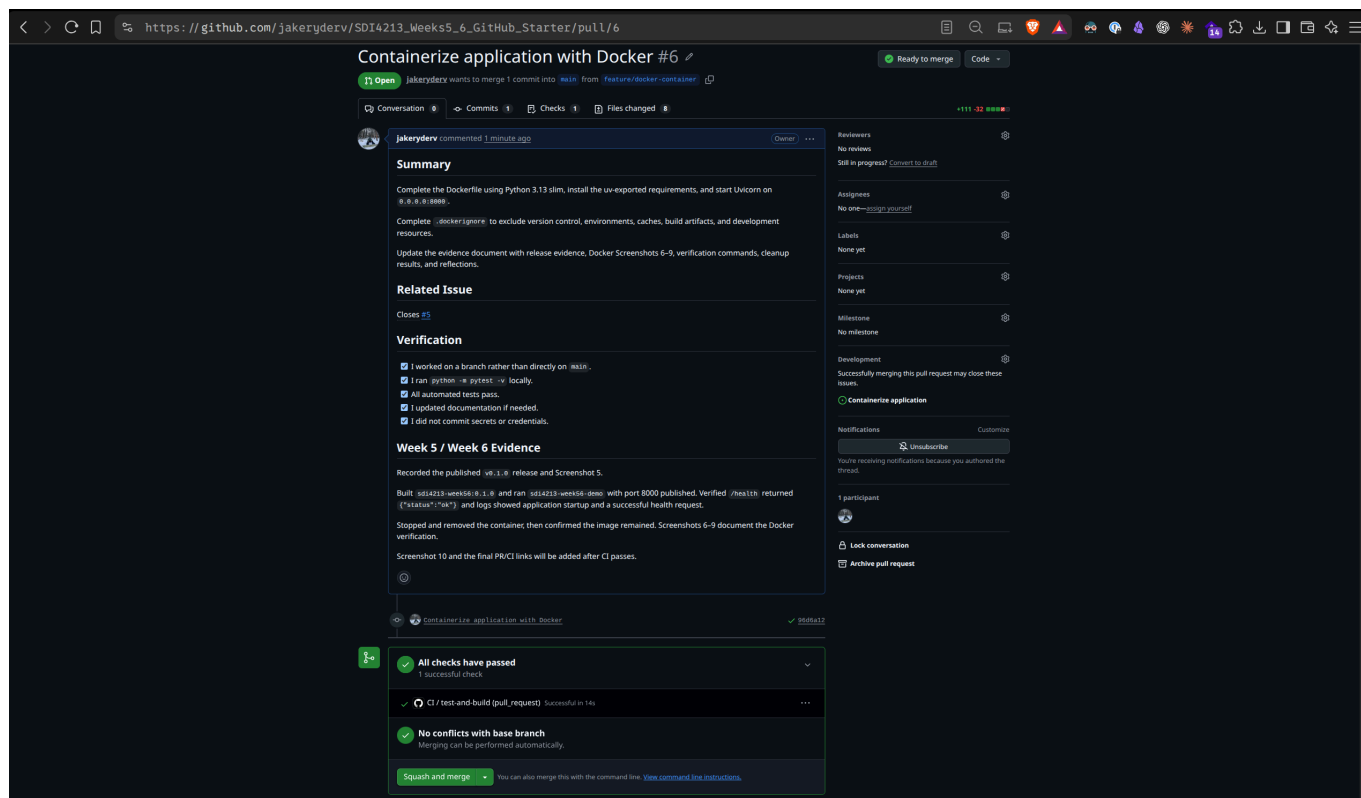
The demo container was stopped and removed after verification:

```
docker stop sdi4213-week56-demo  
docker rm sdi4213-week56-demo  
docker images
```

A subsequent read-only check confirmed that `sdi4213-week56:0.1.0` still exists and no container named `sdi4213-week56-demo` remains. Removing the container did not remove its image.

Final pull request and CI

- Pull request URL: https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/pull/6
- Pull request title: `Containerize application with Docker`.
- Branch: `feature/docker-container` into `main`.
- Successful workflow run URL:
https://github.com/jakeryderv/SDI4213_Weeks5_6_GitHub_Starter/actions/runs/37517840511
- Validated commit: `96d6a129f6a960f20dc5e0e372d265af8a5fd27b`.
- Status at capture: PR #6 was open with all CI checks passing on 2026-10-06, before merge.



Milestone 10 - Containerization changes passed CI through the pull request workflow; the screenshot was captured before merge.

The screenshot shows the repository URL, pull request title, source and target branches, and successful **CI / test-and-build** check.

Reflection

1. What is the difference between a workflow artifact and a Docker image?

A workflow artifact preserves files produced by a CI run. In this exercise, the artifact is a ZIP containing the application source, exported requirements, README, and version file. It still needs a Python environment and dependency installation to run. The Docker image packages the application with the Python runtime, installed dependencies, and startup command so a container can run it. See [GitHub's workflow artifact documentation](#) and [Docker's image explanation](#).

2. Why did you tag the Git release and Docker image with a version?

The Git tag **v0.1.0** identifies the source revision published as the release, while the Docker tag **0.1.0** labels the application's image version. Using the same application version connects the release and image and helps distinguish versions during testing, deployment, and troubleshooting.

3. What does **-p 8000:8000** do?

It maps TCP port 8000 on the host to port 8000 inside the container. The first number is the host port and the second is the container port, allowing **http://localhost:8000/health** to reach Unicorn inside the container. The captured **docker ps** output shows the mapping on IPv4 and IPv6 host addresses. See [Docker's port publishing documentation](#).

4. What would you automate next if this project were moving toward deployment?

I would extend CI to build the Docker image after tests pass, run it temporarily, verify the health endpoint, and remove the test container. After an approved release, the workflow could publish the versioned image to a container registry and deploy it to a staging environment with another health check.